NamLikesToCode

Research Question:

"What customer and service-related factors are associated with the likelihood of customer churn within this organization?"

Goals of the analysis

The goal of this analysis is to develop a multiple logistic regression model that identifies which factors relating to customer and service-related characteristics significantly influence **Churn**. By quantifying the relationship between customers characteristics, demographic attributes and service features, the company can better understand the drivers of churn variability. These insights can be used to improve revenue forecasting, support pricing strategy decisions and inform efforts at improving customer retention.

Assumptions of Logistic Regression:

1. Logistic Regression assumes that the dependent variable is binary when using logistic regression (0/1 or True/False), While ordinal logistic regression requires the dependent variable to be using a scale such as low/medium/high.
2. Logistic regression assumes that observations are independent of one another. This means that each record represents a unique subject/customer, and there are no repeated measurements or matched observations within the dataset.
3. Logistic regression assumes little to no multicollinearity among the independent variables. Highly correlated variables can inflate/amplify standard errors and reduce the reliability of coefficient estimates.
4. Logistic regression assumes a linear relationship between the independent and the log odds(logit) of the outcome variable, rather than a linear relationship between the predictors and outcome itself.

Tool Justification:

Python provides powerful libraries that support efficient data cleaning and preparation. Tools such as Pandas and Numpy makes it possible to create data frames, manipulate data, handle missing values and transform data into a suitable format for logistic regression. These capabilities improve data quality and ensure that the analysis is solid.

Python also offers support for statistical modeling and visualization. Libraries such as matplotlib and seaborn allows for visualization of the data and evaluation.

Additionally, the stats model package provides an easy way to create logistic regression model through its implementation of the **logit()** function, which estimates model parameters using maximum likelihood methods and produces detailed model summaries that support interpretation of coefficients and model performance.

Logistic Regression Justification:

Multiple logistic regression is an appropriate technique for this analysis because the research question examines how multiple independent variables can influence a **binary** dependent variable **Churn**. Unlike linear regression, which is used for continuous outcomes, logistic regression is designed to model the probability that an event occurs. This approach allows the effect of multiple independent variables to be evaluated while controlling for the influence of other predictors. Additionally, logistic regression provides estimates of how each predictor affects the likelihood of churn, supporting the identification of key factors associated with customer retention.

Data cleaning goals and summary:

The primary goal of data cleaning is to provide a high-quality data set with relevant variables for multiple logistic regression involving **Churn** as the response variable. This included ensuring completeness, consistency and appropriate data types for all dependent and independent variables used in the model.

The dataset was reduced to only include variables relevant to the research question, focusing on customer demographics, service characteristics and usage metrics believed to influence **Churn**. This step minimizes noise and improved model interpretability.

The data set is loaded through pd.read_csv with na_values argument to catch null values right at the beginning

```
#Importing Data and removing nulls and nans

filename = 'churn_clean.csv'

df = pd.read_csv(filename, na_values = ['null','N/A',' ', '', '.','nan'])
```

The dataset is then subset to relevant columns

```
#Reducing Dataset to relevant columns for multiple Logistic Regression

df_model = df[

[

   'Churn',

   'Population',

   'Area',

   'Children',

   'Age',

   'Marital',

   'Outage_sec_perweek',

   'Yearly_equip_failure',
```

'Techie',

'Multiple',

'Tenure',

'MonthlyCharge',

'Bandwidth_GB_Year',

'Income',

'Contract'

]]

The data set is then checked for missing values .

#Checking for missing values/nans

print(df_model.isna().sum())

Then checking for data types of the variables.

print(df_model.dtypes)

Converted Categorical columns from Object to Category

#convert categorical columns into the correct datatype

cat_cols = ['Area',

'Marital',

'Techie',

'Multiple',

'Contract']

for col in cat_cols:

df_model[col] = df_model[col].astype('category')

Binary columns values are replaced from Yes/No to 1/0

#Turn binary categoricals into Yes/No for easier model interpretation

df_model['Churn'] = df_model['Churn'].replace({'Yes':1,'No':0})

df_model['Techie'] = df_model['Techie'].replace({'Yes':1,'No':0})

df_model['Multiple'] = df_model['Multiple'].replace({'Yes':1,'No':0})

Checking to see if the variables have the correct values by using **unique()** method

#Checking to see if both column have been correctly transformed to 0/1 for better interpretation of the Logistic regression model

print(df_model[['Techie','Churn','Multiple']].head())

print(df_model['Techie'].unique())

print(df_model['Churn'].unique())

print(df_model['Multiple'].unique())

Checking other categorical columns to see if it has proper observations.

print(df_model['Area'].unique())

print(df_model['Marital'].unique())

print(df_model['Multiple'].unique())

print(df_model['Contract'].unique())

Categorical binary variables that are set as categorical are now set as Int for better model interpretation.

```
#Turn back categorical binaries into Integer for better model interpretation

df_model['Techie'] = df_model['Techie'].astype(int)

df_model['Multiple'] = df_model['Multiple'].astype(int)
```

Continuous variables are then checked for potential outliers, using **sns.boxplot.**

```
sns.boxplot(x='Population', data=df_model)

sns.boxplot(x='Children',data=df_model)

sns.boxplot(x='Age',data=df_model)

sns.boxplot(x='Outage_sec_perweek', data=df_model)

sns.boxplot(x='Yearly_equip_failure',data=df_model)

print((df_model['Yearly_equip_failure']> 2).sum())

sns.boxplot(x='Tenure',data=df_model)

sns.boxplot(x='MonthlyCharge',data=df_model)

sns.boxplot(x= 'Bandwidth_GB_Year',data=df_model)

sns.boxplot(x='Income',data=df_model)
```

The Income variable exhibited significant right skew due to a small number of extreme values. To reduce the influence of outliers while preserving all observations, income values above $175,000 were capped at $175,000.

```
#Capping max values at 175k

df_model.loc[df_model['Income'] > 175000, 'Income'] = 175000
```

While variables Children and Population also carries outliers, both variables kept their outliers because Population’s outliers are a reflection of larger urban areas and Children’s outliers were rare but still possible, with 401 observations having over 7 children.

print((df['Children']>7).sum())

Describe the data variables:

Churn is a categorical binary variable representing the customer if they have churned within the last month. The variable shows most customers have not churned within the last month at 7350, 73.5% of the column while customers who have churned are at 2650 observations, 26.5% of the column.

```
Churn
0    7350
1    2650
Name: count, dtype: int64
Churn
0    73.5
1    26.5
Name: proportion, dtype: float64
```

Independent Variables:

Population:

Population is a discrete numerical variable representing the population within a mile radius of the customer. The mean of Population is 9756.56, with a median of 2910.5. The lowest record of Population is 0 while the largest record is 111850, with a range of 111850. The standard deviation of Population is 14432.69. The distributions first quartile is 738, the third quartile is 13168. The shape of the distribution is unimodal with a strong right skew due to high population density, suggesting urban metropolitan cities pull the mean up higher as the mean is much larger than the median.

```
count     10000.000000
mean       9756.562400
std       14432.698671
min           0.000000
25%         738.000000
50%        2910.500000
75%       13168.000000
max      111850.000000
Name: Population, dtype: float64
Range: 111850
```

Area:

Area is a nominal categorical variable representing the customers area demographic. The column has values of **Suburban, Rural** and **Urban.** The dataset shows that most customers are in suburban areas at 33.46%, rural areas at 33.27%, urban areas at 33.27%. This variable is relevant to **Churn** as urban areas may have different internet/phone/tv providers who have more competitive pricing

```
Area
Suburban    3346
Rural       3327
Urban       3327
Name: count, dtype: int64
Area
Suburban    33.46
Rural       33.27
Urban       33.27
Name: proportion, dtype: float64
```

Children:

Children is a discrete numerical variable representing the number of children in the customers household at the time of sign up. The mean of Children is 2.08, with a median of 1. The lowest record for Children is 0 and the largest is 10 with a range of 10. Children has a standard deviation of 2.14, with the first quartile at 0 and third quartile at 3. The distribution is shaped as a unimodal with a strong right skew due to outliers.

```
count    10000.0000
mean         2.0877
std          2.1472
min          0.0000
25%          0.0000
50%          1.0000
75%          3.0000
max         10.0000
Name: Children, dtype: float64
Range: 10
```

Age:

Age is a continuous numerical variable representing the customers age at the time of service sign up. The mean of Age is 53, with a median of 53. The lowest record for Age is 18 while the highest record is 89. Age has a standard deviation of 20.69 with first quartile at 35 and third quartile at 71 with a range of 71

```
count    10000.000000
mean        53.078400
std         20.698882
min         18.000000
25%         35.000000
50%         53.000000
75%         71.000000
max         89.000000
Name: Age, dtype: float64
Range: 71
```

Marital:

Marital is a nominal categorical variable representing the marital status of the customer as reported in sign-up information. Marital has 5 possible values from most to least with 'Divorced' at 2092 observations, 20.92%, 'Widowed' at 2027 observations, 20.27%, 'Separated' at 2014 observations, 'Never Married' at 1956 observations, 19.56%, lastly 'Married' at 1911, 19.11%.

```
Marital
Divorced         2092
Widowed          2027
Separated        2014
Never Married    1956
Married          1911
Name: count, dtype: int64
Marital
Divorced         20.92
Widowed          20.27
Separated        20.14
Never Married    19.56
Married          19.11
Name: proportion, dtype: float64
```

Outage_sec_perweek:

The Outage_sec_perweek a continous numerical variable which represents the number of seconds per week of system outages in the customers neighborhood. The mean is 10 seconds, with a median of 10.01, indicating a symmetrical distribution. Standard deviation is 2.97 seconds, with a minimum value of 0.09, and maximum value of 21.2 seconds. First quartile is at 8.01 seconds while third quartile is at 11.97 seconds. Lastly with a range of 21.1.

```
count    10000.000000
mean        10.001848
std          2.976019
min          0.099747
25%          8.018214
50%         10.018560
75%         11.969485
max         21.207230
Name: Outage_sec_perweek, dtype: float64
Range: 21.10748306
```

Yearly_equip_failure:

Yearly_equip_failure is a discrete numerical variable that represents how often in the past year, the customers equipment has failed and required a reset/replacement.

The mean of Yearly_equip_failure is at .39, with a median of 0, indicating that most customers do not have problems with their equipment. Minimum value is at 0 while the maximum value is at 6. First quartile is at 0 while third quartile is at 1. The standard deviation is .63, with a range of 6.

```
count    10000.000000
mean         0.398000
std          0.635953
min          0.000000
25%          0.000000
50%          0.000000
75%          1.000000
max          6.000000
Name: Yearly_equip_failure, dtype: float64
Range: 6
```

Techie:

Techie is a binary categorical variable that represents if the customer is technically inclined.

The column shows most customers do not consider themselves to be technically inclined at 8321 observations, 83.21% with 1679 consider themselves to be technically inclined at 16.79%.

```
Techie
0    8321
1    1679
Name: count, dtype: int64
Techie
0    83.21
1    16.79
Name: proportion, dtype: float64
```

Multiple:

Multiple is a binary categorical variable that represents if the customer has multiple phone lines for their phone service. With 5392 observations at 53.92% not having multiple phone lines while 4608 at 46.08% do having multiple phone lines.

```
Multiple
0    5392
1    4608
Name: count, dtype: int64
Multiple
0    53.92
1    46.08
Name: proportion, dtype: float64
```

Tenure:

Tenure is a continuous numerical variable representing the number of months the customer has stayed with this service provider. The mean of Tenure is approximately 34.52 months, with a median of 35.43 months. The lowest Tenure was 1 month while the highest tenure is 71.99 months, with a range of 70.99 months. The standard deviation of Tenure is 26.44 months.

The distributions first quartile is 7.91 months, the third quartile at 61.47 months. The distribution is bimodal, suggesting two customer groups with differing tenures.

```
count    10000.000000
mean        34.526188
std         26.443063
min          1.000259
25%          7.917694
50%         35.430507
75%         61.479795
max         71.999280
Name: Tenure, dtype: float64
Range: 70.99902066
```

MonthlyCharge:

MonthlyCharge is a continuous numerical variable representing the amount in dollars for the customer's monthly bill. The mean of MonthlyCharge is approximately $172.62, with a median of $167,48. MonthlyCharge values range from $79.97 to $290.16, resulting in a range of $210.

The standard deviation of $42.94 suggest moderate variability in monthly pricing across customers, reflecting difference in service usages, additional choices in plans and features.

First quartile is $139.97 and third quartile is $200.73. The distribution is unimodal.

```
count    10000.000000
mean       172.624816
std         42.943094
min         79.978860
25%        139.979239
50%        167.484700
75%        200.734725
max        290.160419
Name: MonthlyCharge, dtype: float64
Range: 210.181559
```

Bandwidth_GB_Year:

Bandwidth_GB_Year is a continuous numerical variable representing a customer's annual data usage. The mean of annual bandwidth usage is approximately 3,392 gb, with a median of 3,280 gb, indicating a relatively symmetric distribution. Bandwidth usage ranges from approximately 156 gb to 7,159 gb, resulting in a range of about 7,003 gb. The standard deviation of 2185 suggest variability in data usage among customers, which signifies it could be a strong potential predictor of MonthlyCharge. The first quartile is 1236.47gb and the third quartile is 5586.14 gb. The distribution is bimodal with two distinct peaks, suggesting two customer groups with differing levels of bandwidth usage.

```
count    10000.000000
mean      3392.341550
std       2185.294852
min        155.506715
25%       1236.470827
50%       3279.536903
75%       5586.141370
max       7158.981530
Name: Bandwidth_GB_Year, dtype: float64
Range: 7003.4748152
```

Income:

Income is a continuous numerical variable representing the customers income at time of signup. Income has a mean of $39,768.59, with a maximum income capped at $175,000 and a minimum income of $348.67, with a range of $174.651. Income's median is at $33,170.6, with a standard deviation of $27, 976, first quartile at $19,224.71 and third quartile at $53,246. With the mean being higher than median, it suggest a slight skew to the right due to higher income outliers that are capped.

```
count     10000.000000
mean      39768.591427
std       27976.727587
min         348.670000
25%       19224.717500
50%       33170.605000
75%       53246.170000
max      175000.000000
Name: Income, dtype: float64
Range: 174651.33
```

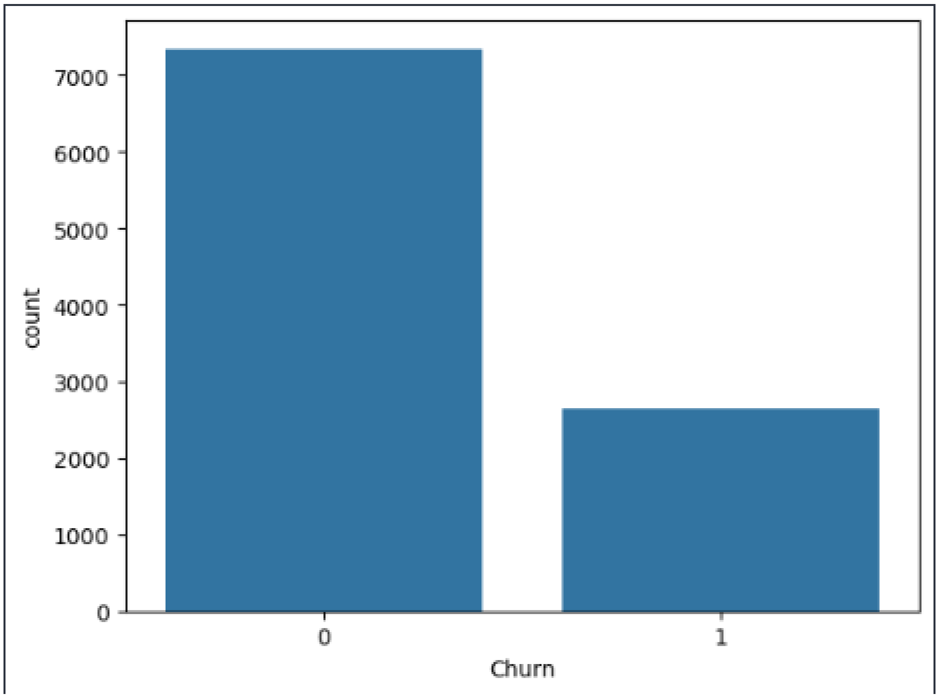
Contract:

Contract is a nominal categorical variable describing the type of contracts the customer has chosen for their service. The most common observation of Contract is Month-to-Month at 5456 observations, at 54.56%. The second most common observation is **Two Year** contracts at 2442 observations at 24.42%. The least common observation is **One Year** at 2102 observations, 21.02% of the distribution.

```
Contract
Month-to-month    5456
Two Year          2442
One year          2102
Name: count, dtype: int64
Contract
Month-to-month    54.56
Two Year          24.42
One year          21.02
Name: proportion, dtype: float64
```

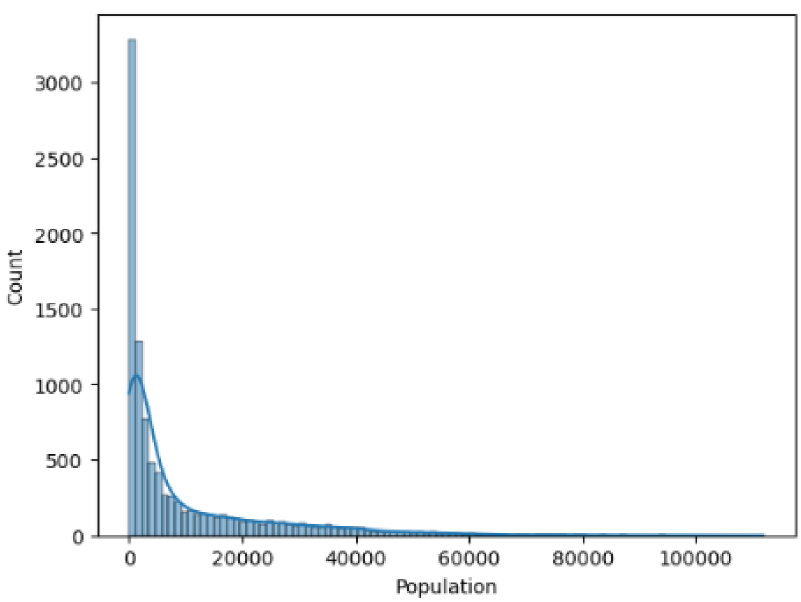
Visualizations of the variables:

Churn:



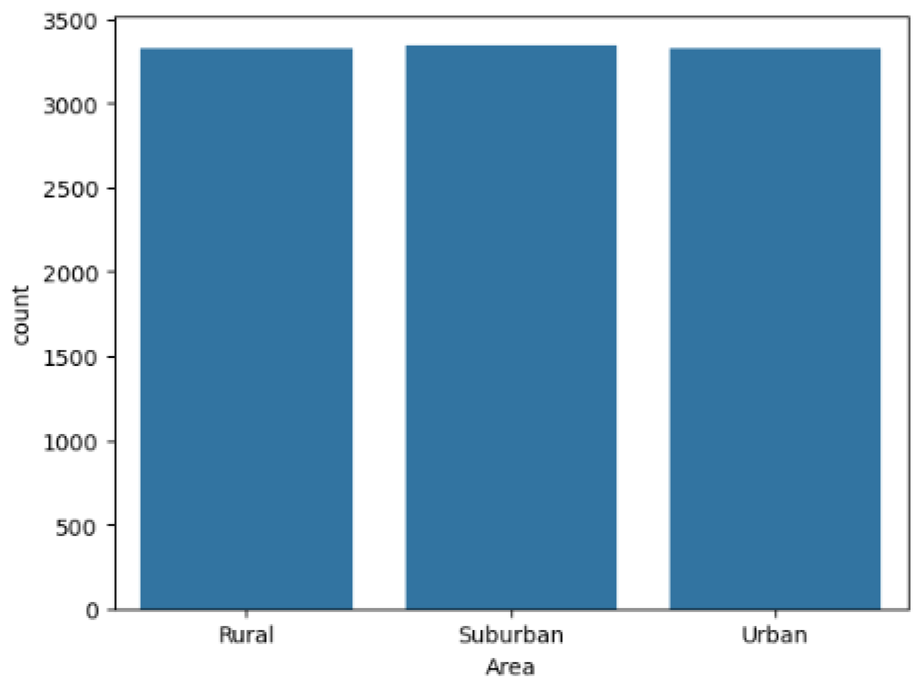
Churners show to be around 25% of the distribution out of the 10k rows.

Population:



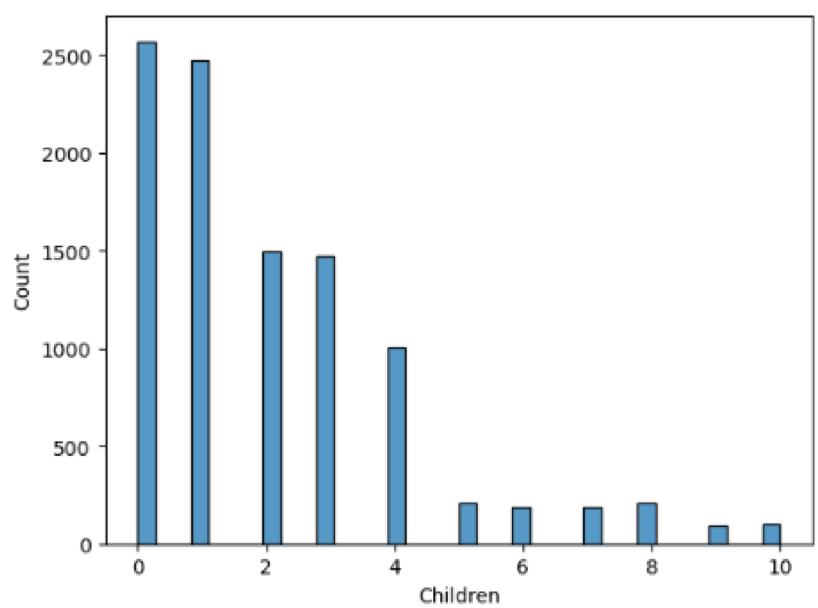
Population shows to be a unimodal distribution with a strong right skew due to outliers of customer who live in urban areas while most customers live in areas with fewer people around their location.

Area:



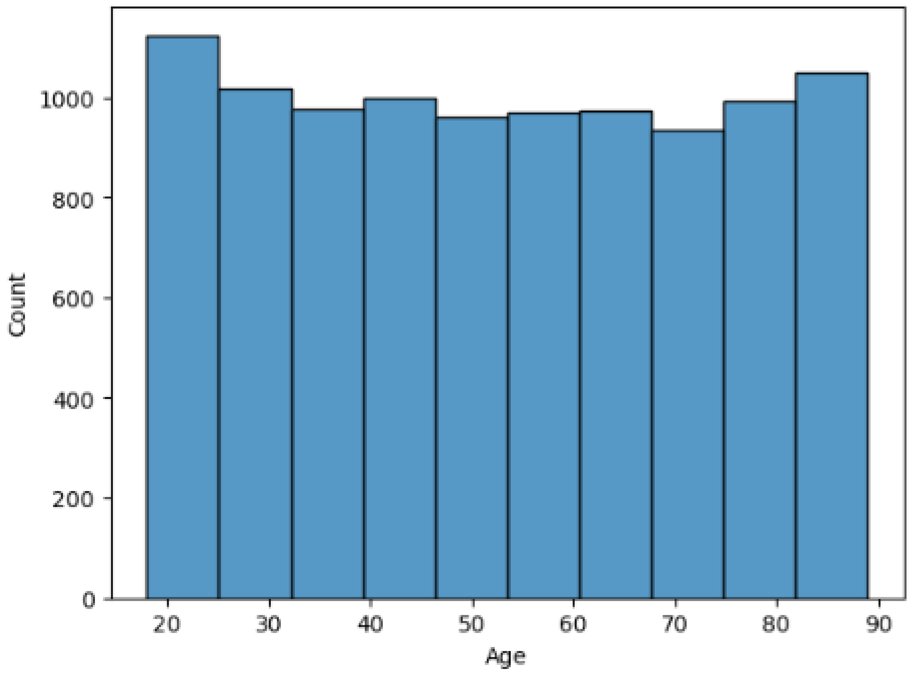
Area shows that customers are evenly split by area type.

Children:



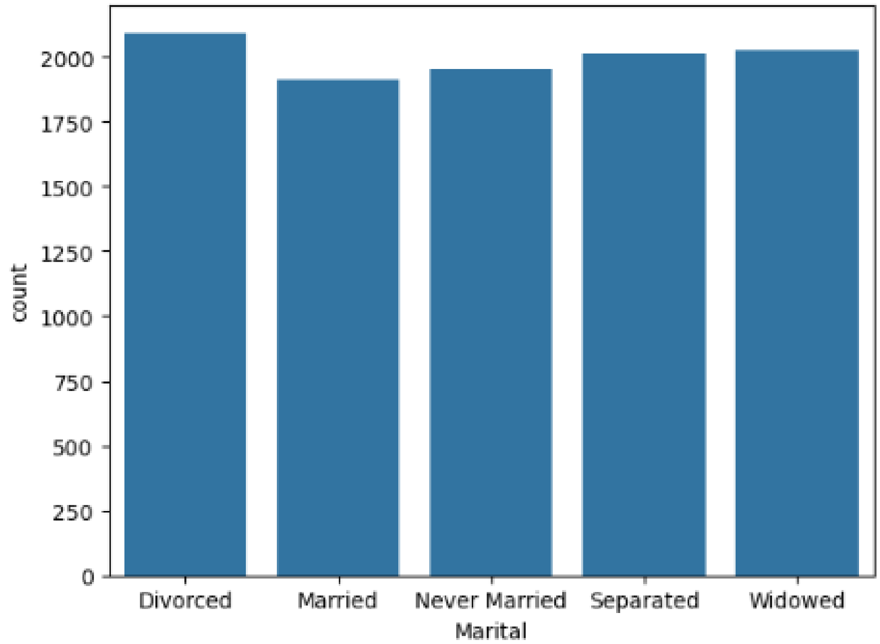
Children shows that most customers do not have children, the distribution is unimodal with a right skew with a large drop starting from 4 children.

Age:



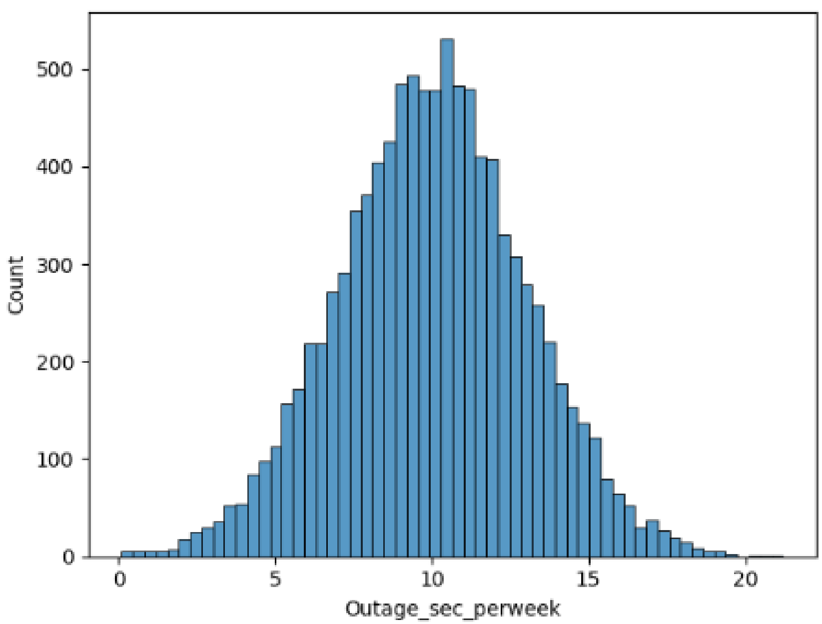
Uniform distribution with 18 to 25 being the most common customer ages slightly compared to the rest of the demographics.

Marital:



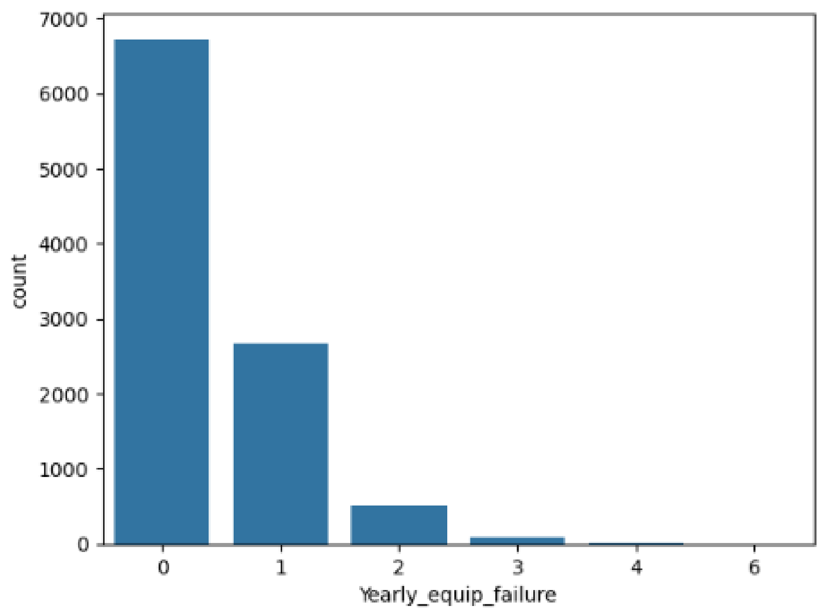
Uniform distribution with divorced being the most common and married being the least common.

Outage_sec_perweek:



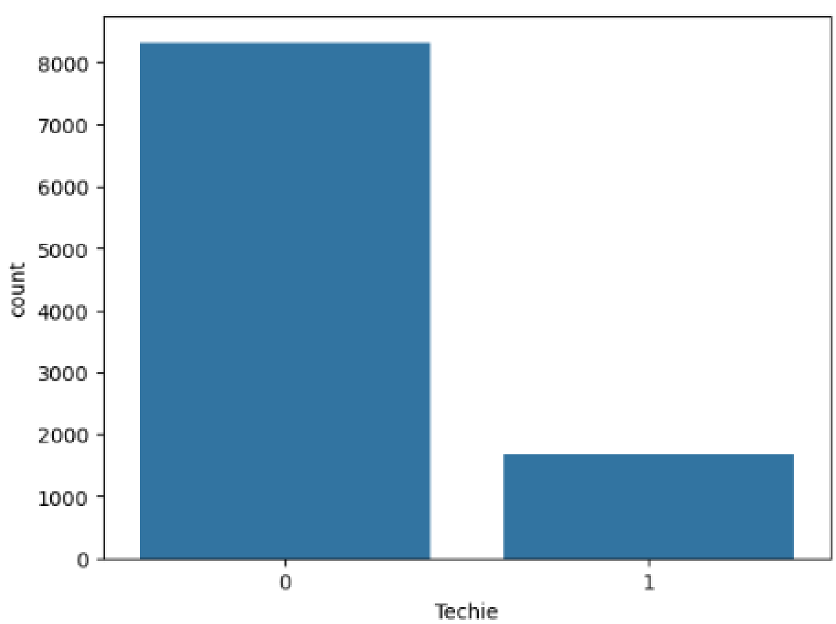
Normal Distribution, most customers have outage for 11/12 seconds per week.

Yearly_equip_failure:



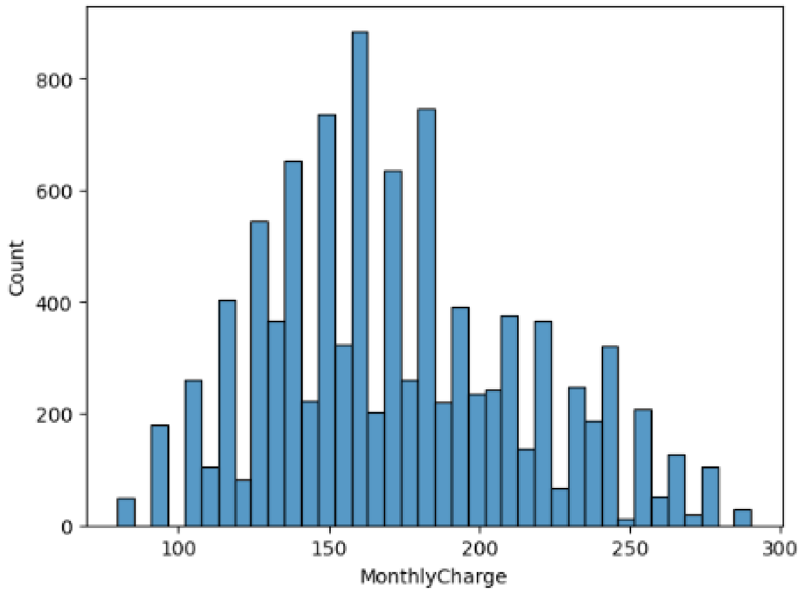
Unimodal distribution showing that most customers do not have equipment failure that needed to be replaced/reset in the past year. Far right skew due to outliers.

Techie:



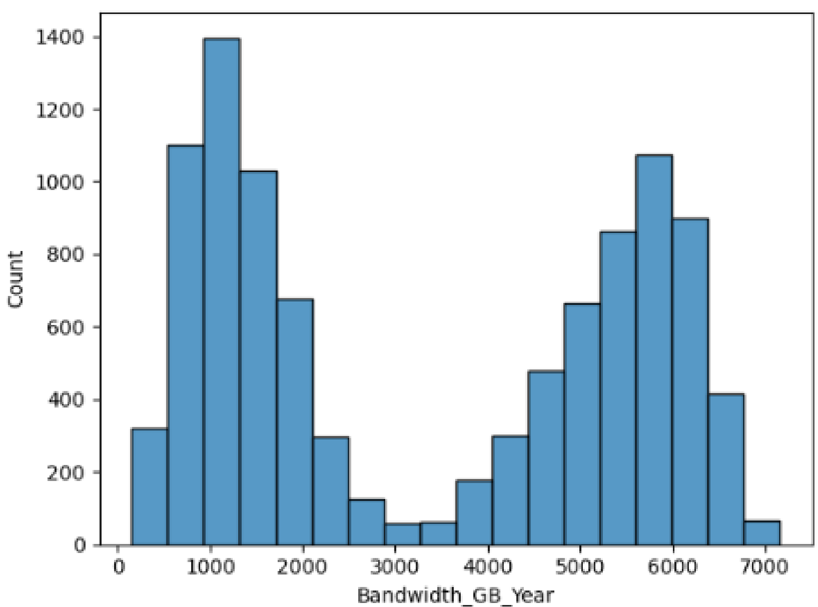
Customers who declare not to be technically inclined is the majority at slightly over 8000 of the distribution.

MonthlyCharge:



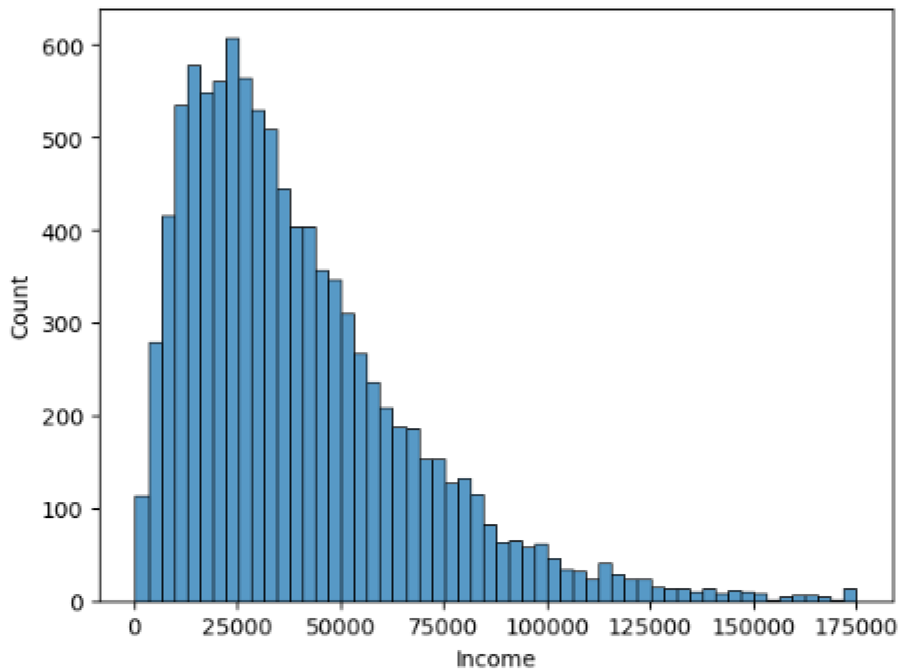
MonthlyCharge has a normal distribution, with most customers paying at around 160-170 dollars per month.

Bandwidth_GB_Year:



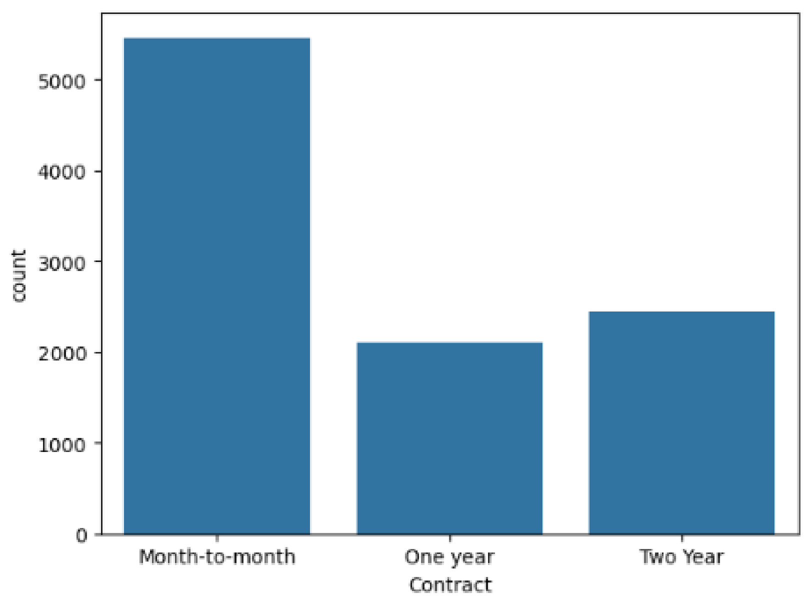
Bandwidth_GB_Year has bimodal distribution, showing two significant groups of customers with different usage, one at around 1000gb usage yearly and the other at around 5800/6000gb.

Income:



Income is a unimodal distribution with a strong right skew due to higher income earners that has been capped.

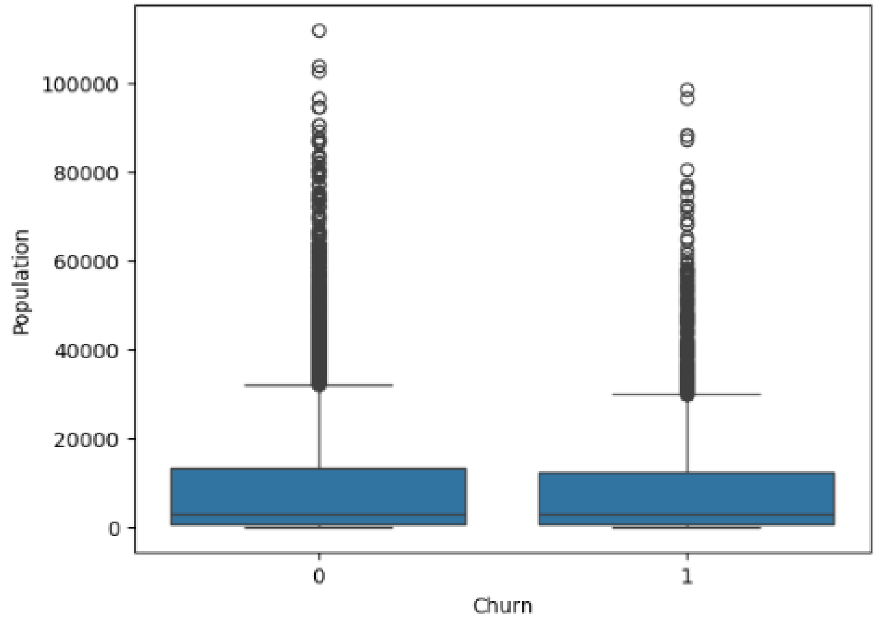
Contract:



Month-to-month is the most popular contract followed by Two Year then lastly One Year contracts.

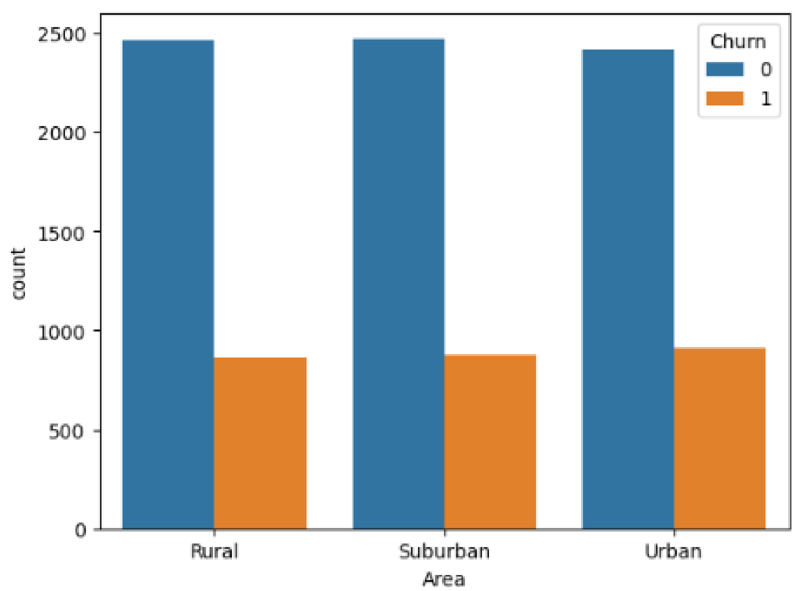
Bivariate graphs:

Churn X Population



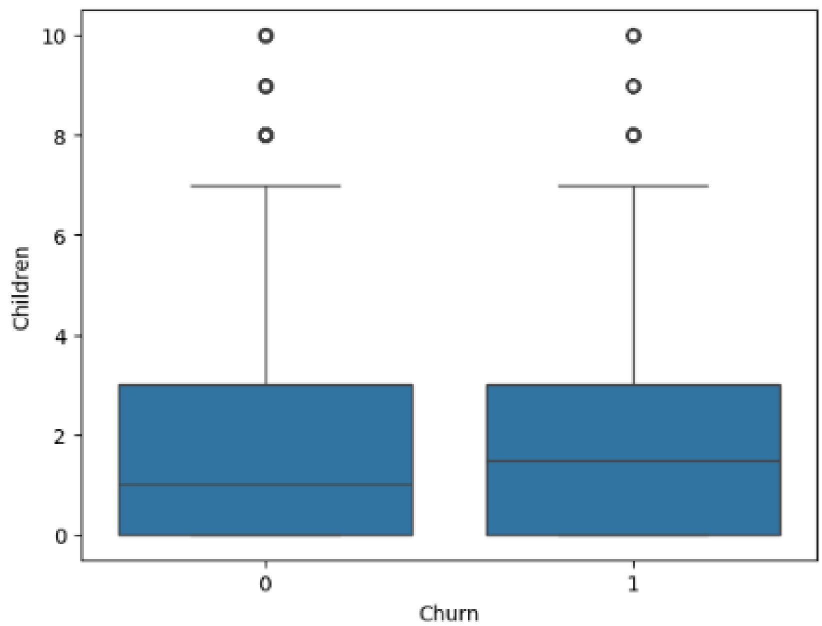
Median, Q1, Q3 and whiskers are highly similar for both churners and non churners.

Churn x Area



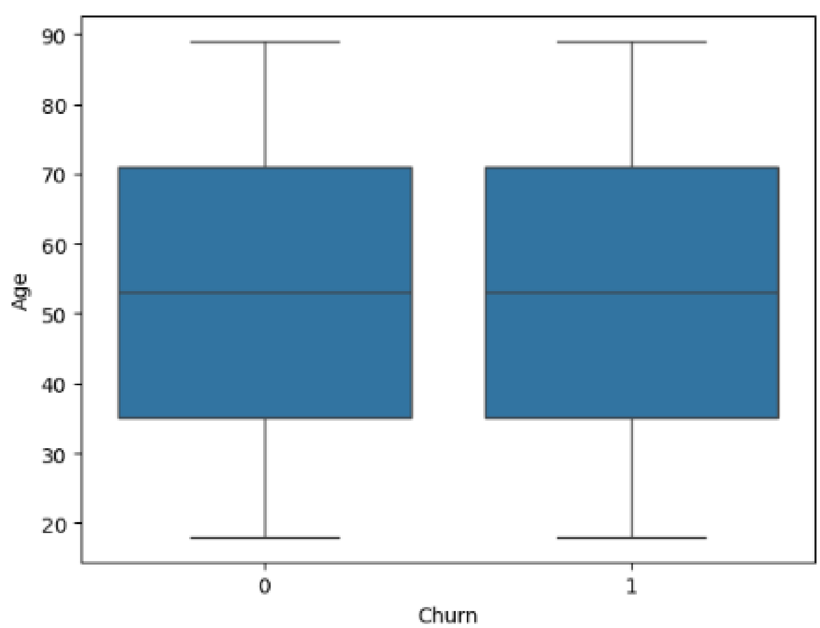
Distribution ratio for churners and non-churners in all 3 residential types are very similar as well as the residential distribution of the population.

Churn x Children



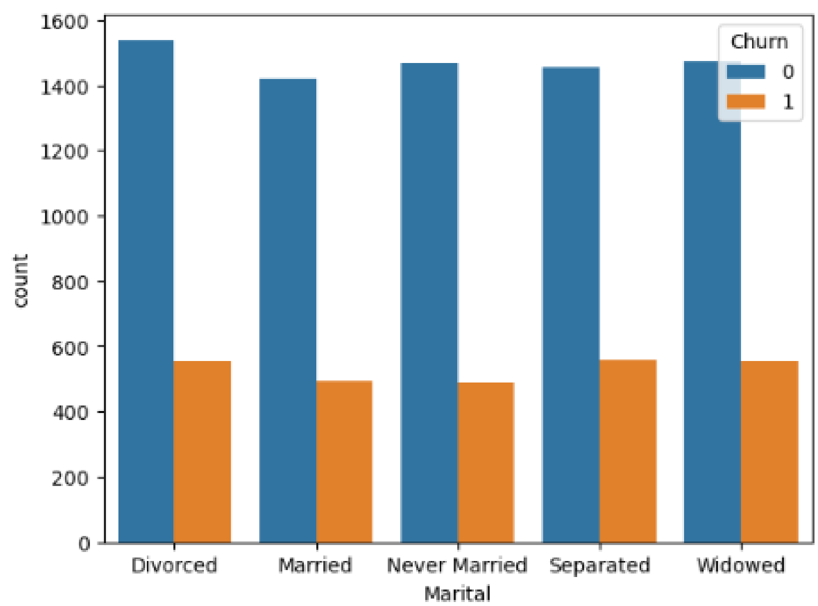
Q1 and Q3 are very similar for churners and non-churners. Whiskers extend to similar values, outliers exist in both sides and are in similar numbers of children, median for churners are slightly higher.

Churn x Age



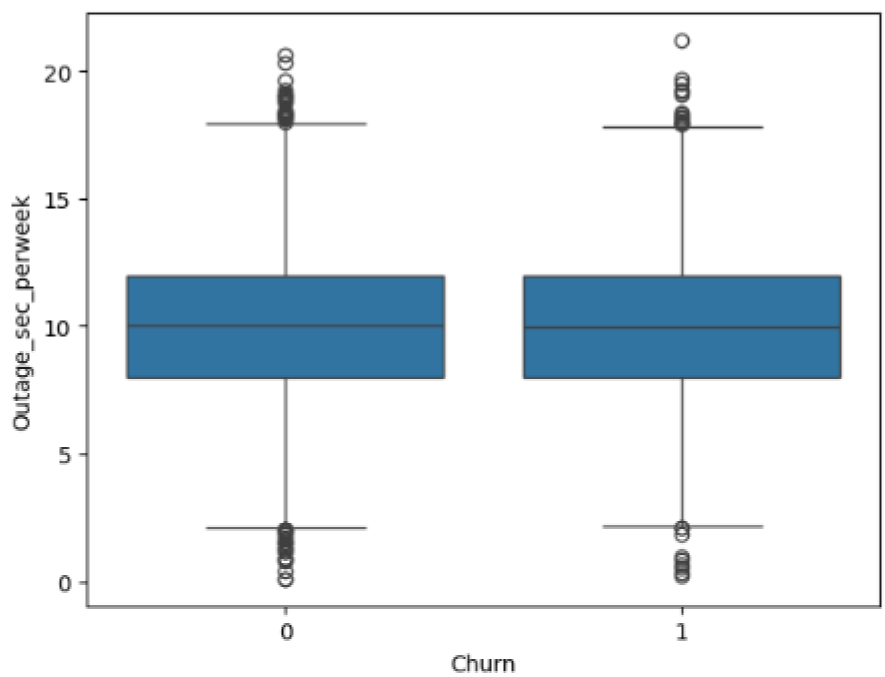
Both boxplots show to be very similar.

Churn x Marital



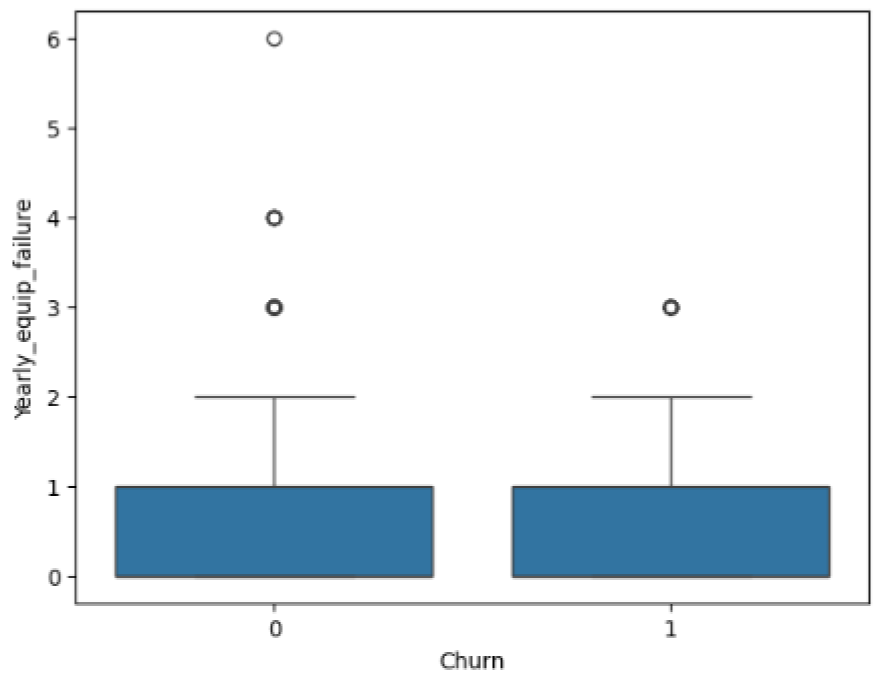
All groups have a similar ratio of non-churners being over twice the size of churners in all relationship status.

Churn x Outage_sec_perweek



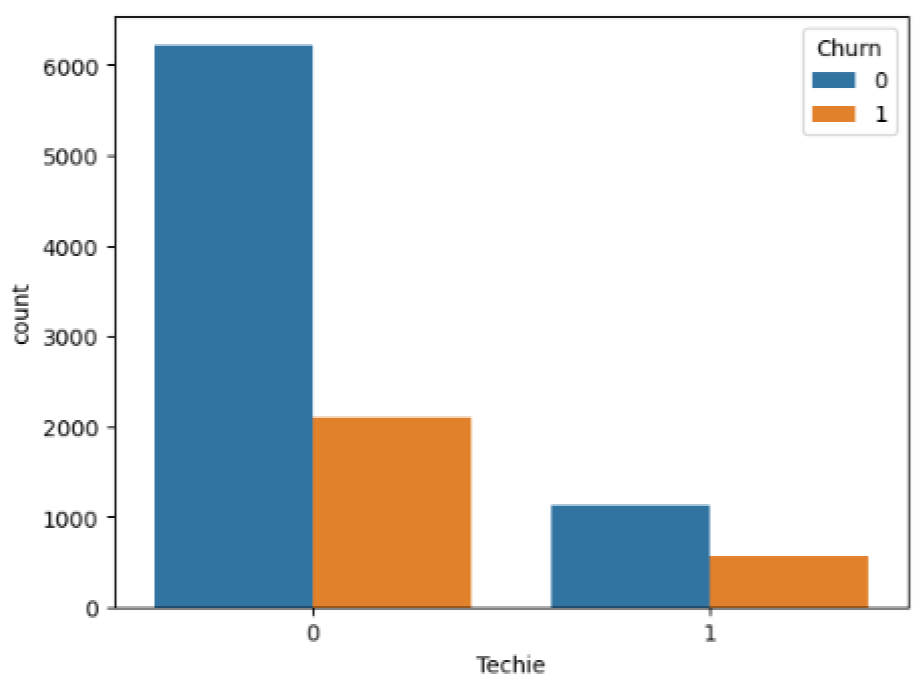
Both distributions are very similar, same median/Q1/Q3. Outliers have little difference. Both upper and lower whiskers are very similar.

Churn x Yearly_equip_failure



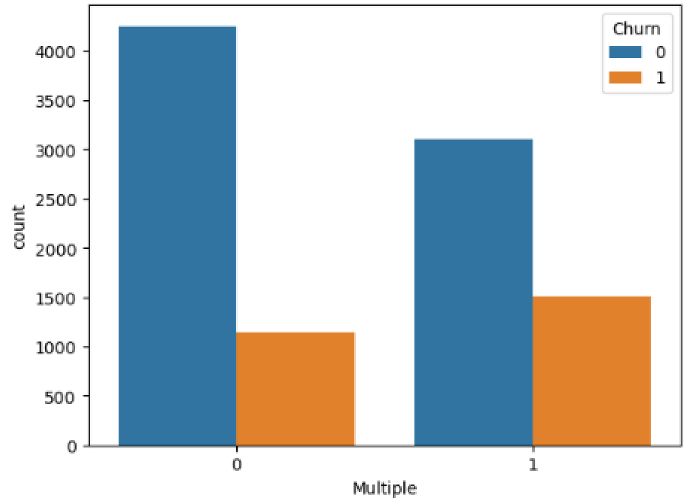
Churners and non-churners have similar distribution below the upper whisker, while non churners have higher and more outliers in terms of equipment failure.

Churn x Techie



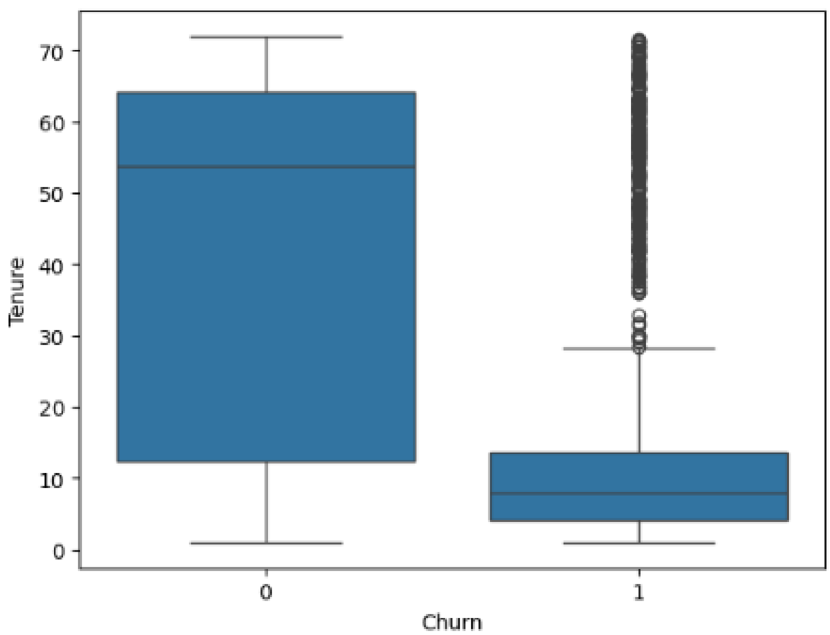
Non-techies have a larger ratio of non-techies to techies for non-churners while churners have a 2 to 1 ratio of non techie to techies.

Churn x Multiple



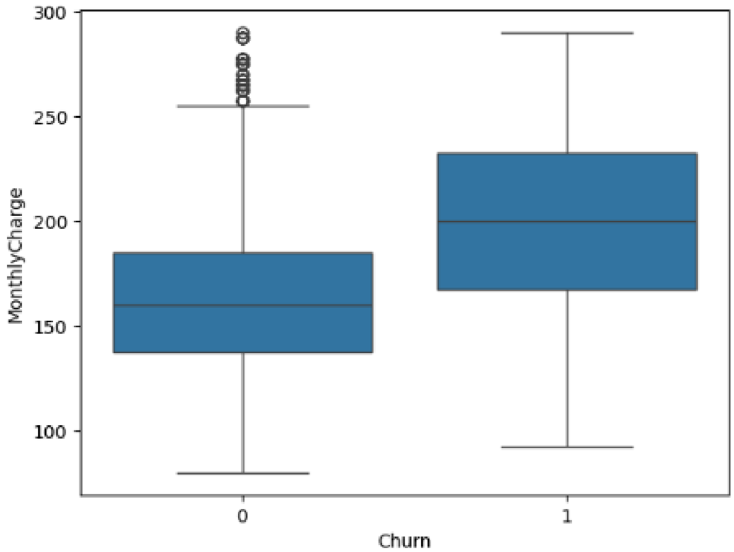
Non-Churners have a higher ratio of customers without multiple lines to does with multiple lines. While Churners have a 2:1 ratio of those without multiple lines to those who do.

Churn x Tenure



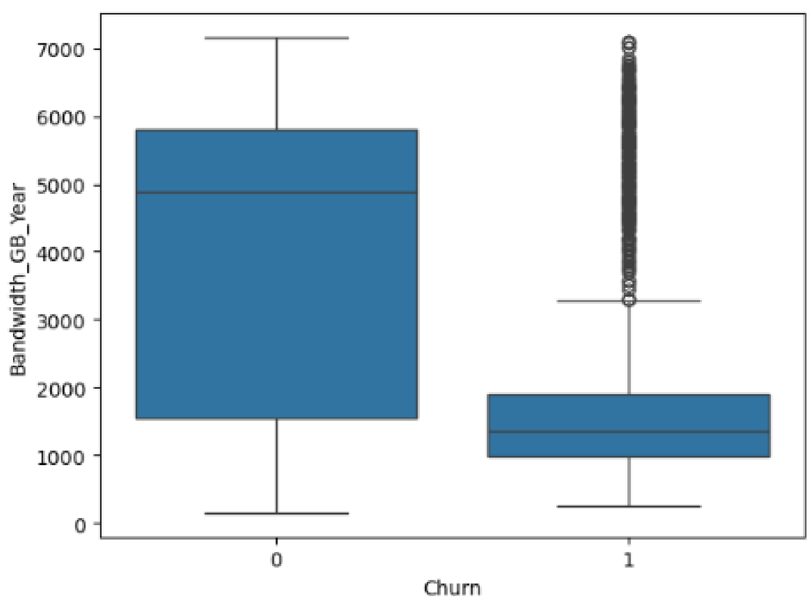
Churned customers show much lower median/Q1/Q3/Upper whisker compared to non-churner customers in Tenure length. Churners show more present outliers.

Churn x MonthlyCharge



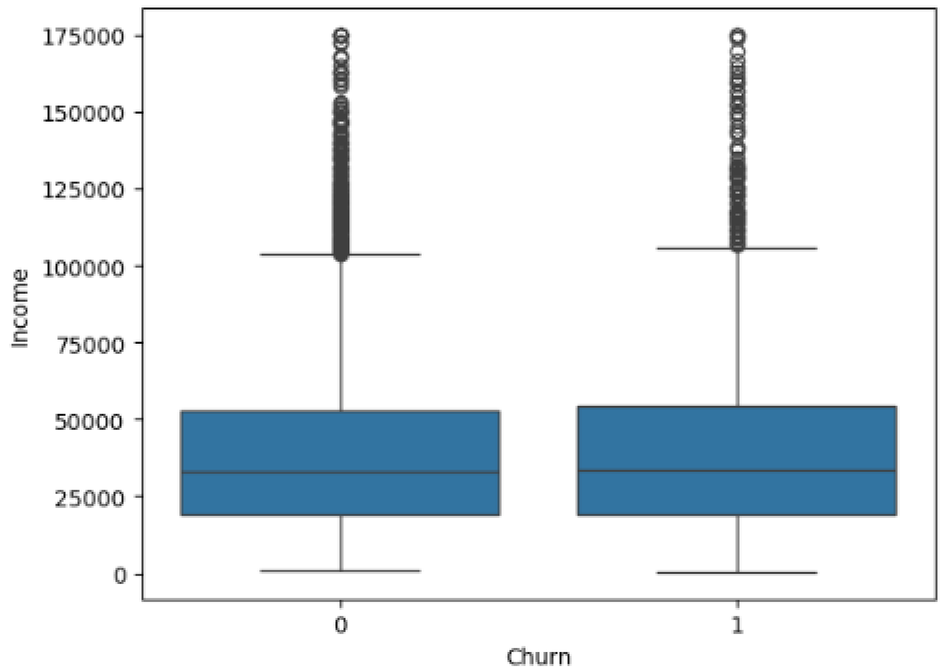
Churners have a higher median/Q1/Q3/Lower and upper whisker in MonthlyCharge while non churners have more frequent outliers above the upper whiskers.

Churn x Bandwidth_GB_Year



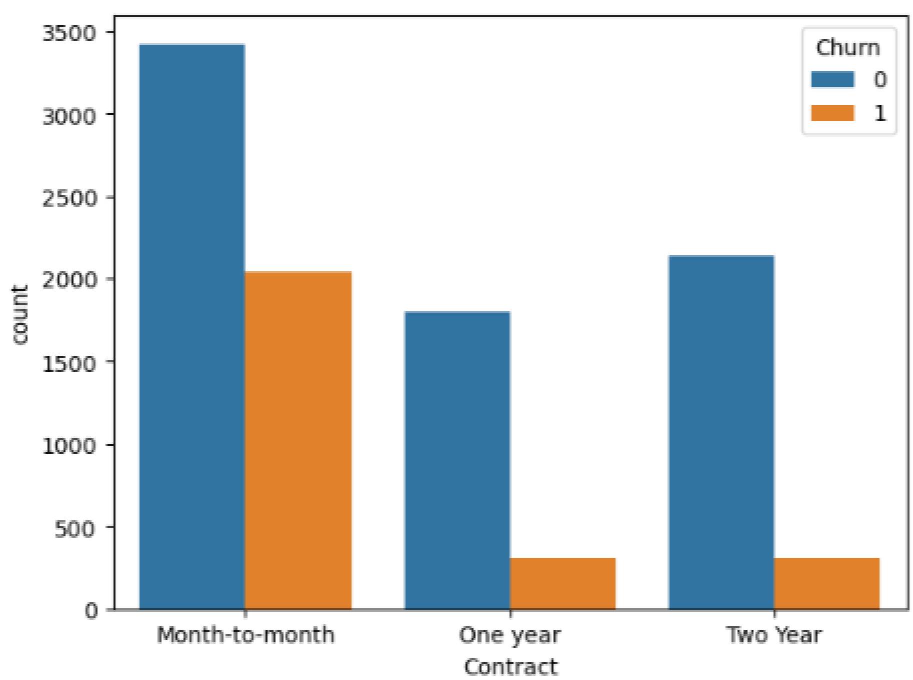
Churners show to use less bandwith per year outside of outliers compared to non churners. With a lower median/Q1/Q3/Upper whisker compared to nonchurners. While non-churners median is significantly higher than churners.

Churn x Income



Both distributions show to be very similar.

Churn x Contract



For contracts that are a year or longer, Churn is drastically reduced.

Data transformation for preparation of creating Logistic Regression model summary:

The goals of the data transformation were to prepare the dataset for multiple logistic regression analysis using **LOGIT** from statsmodels.formula.api. Because logistic regression requires all predictors variables to be numeric and free of multicollinearity, categorical variables were transformed into appropriate numeric representations to ensure compatibility with the model and to allow for meaningful interpretations of coefficients in terms of odds of churn.

Several categorical variables in this dataset were binary in nature (Yes/No values). These variables were converted into numeric indicator variables where Yes = 1 and No = 0. This transformation allows the estimated logistic regression coefficients to be directly interpreted as changes in the log-odds of customer churn associated with the presence of each feature.

```
#convert categorical columns into the correct datatype

cat_cols = ['Area',

    'Marital',

    'Techie',

    'Multiple']

for col in cat_cols:

  df_model[col] = df_model[col].astype('category')


#Turn binary categoricals into Yes/No for easier model interpretation

df_model['Churn'] = df_model['Churn'].replace({'Yes':1,'No':0})

df_model['Techie'] = df_model['Techie'].replace({'Yes':1,'No':0})

df_model['Multiple'] = df_model['Multiple'].replace({'Yes':1,'No':0})

#Turn back categorical binaries into Integer for better model interpretation


df_model['Techie'] = df_model['Techie'].astype(int)

df_model['Multiple'] = df_model['Multiple'].astype(int)
```

Categorical nominal variables with more than two categories (Area/Marital/Contract) were converted using one-hot encoding with **getdummies()**. For each variable, one category was dropped to serve as the reference group (k-1 = number of columns -1), preventing multi-collinearity/perfect linearity in the regression model.

```
#One hot encode categorical nominal categories


one_hot_category= ['Marital','Area','Contract']


df_model=pd.get_dummies(df_model,
                    columns=one_hot_category,
                    drop_first=True
                   )
```

Model Results:

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                  Churn   No. Observations:                10000
Model:                          Logit   Df Residuals:                     9980
Method:                           MLE   Df Model:                           19
Date:                Sat, 24 Jan 2026   Pseudo R-squ.:                  0.5930
Time:                        07:29:29   Log-Likelihood:                -2353.3
converged:                       True   LL-Null:                       -5782.2
Covariance Type:            nonrobust   LLR p-value:                     0.000
=================================================================================================
                                   coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------------------
Intercept                       -7.0768      0.286    -24.771      0.000      -7.637      -6.517
Marital_Married[T.True]          0.0730      0.117      0.624      0.533      -0.156       0.302
Marital_Never_Married[T.True]   -0.0391      0.117     -0.335      0.738      -0.268       0.190
Marital_Separated[T.True]        0.1500      0.116      1.297      0.195      -0.077       0.377
Marital_Widowed[T.True]          0.2853      0.115      2.476      0.013       0.059       0.511
Area_Suburban[T.True]           -0.0353      0.091     -0.387      0.699      -0.214       0.144
Area_Urban[T.True]               0.0262      0.091      0.289      0.772      -0.151       0.204
Contract_One_year[T.True]       -3.1580      0.119    -26.499      0.000      -3.392      -2.924
Contract_Two_Year[T.True]       -3.2583      0.117    -27.782      0.000      -3.488      -3.028
Population                   -5.735e-07   2.59e-06     -0.222      0.825   -5.64e-06     4.5e-06
Children                        -0.1133      0.018     -6.208      0.000      -0.149      -0.078
Age                              0.0145      0.002      7.590      0.000       0.011       0.018
Outage_sec_perweek              -0.0006      0.012     -0.048      0.962      -0.025       0.024
Yearly_equip_failure            -0.0187      0.058     -0.321      0.748      -0.133       0.095
Techie                           0.9685      0.098      9.901      0.000       0.777       1.160
Multiple                         0.0914      0.078      1.167      0.243      -0.062       0.245
Tenure                          -0.4527      0.016    -27.840      0.000      -0.485      -0.421
MonthlyCharge                    0.0352      0.001     27.884      0.000       0.033       0.038
Bandwidth_GB_Year                0.0042      0.000     23.151      0.000       0.004       0.005
Income                        6.655e-07   1.32e-06      0.505      0.613   -1.92e-06    3.25e-06
=================================================================================================
```

Backward stepwise elimination was employed as a statistically based variable removal procedure. The initial logistic regression model included all theoretically relevant prediction variables. At each iteration, predictors with the highest p-values were evaluated for removal and the model is to be refitted.

Model comparison will be guided by AIC(Akaike Information Criterion), which evaluates model fit while penalizing unnecessary complexity and low contribution variables. AIC is appropriate for logistic regression models, as a decrease in AIC indicates improved explanatory performance without overfitting. The BIC (Bayesian Information Criteria) is also reviewed as a supporting metric to assess model performance, as a lower value indicates a better balance between fitness and complexity.

Additional model evaluation metrics were examined to confirm overall model performance. Log-likelihood values were used to assess model fit, with higher values indicating improved fit. Pseudo R-squared was used to evaluate the relative improvement of the fitted model compared to a null (Intercept-only) model, where higher values indicate stronger explanatory power. The likelihood ratio test (LLR p-value) was used to determine whether the improvement over the null model was statistically significant, with lower p-values indicating stronger evidence that the predictors collectively contribute to explaining customer churn.

Initial Model:

```
#Creating initial logistic regression model

model1= logit("Churn ~ Population + Children + Age + Outage_sec_perweek + Yearly_equip_failure + Techie + Multiple + Tenure + MonthlyCharge"

        "+ Bandwidth_GB_Year + Income + Marital_Married + Marital_Never_Married + Marital_Separated + Marital_Widowed + Area_Suburban"

        "+ Area_Urban + Contract_One_year + Contract_Two_Year" ,data=df_model).fit()
```

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                  Churn   No. Observations:                10000
Model:                          Logit   Df Residuals:                     9980
Method:                           MLE   Df Model:                           19
Date:                Sat, 24 Jan 2026   Pseudo R-squ.:                  0.5930
Time:                        09:02:35   Log-Likelihood:                -2353.3
converged:                       True   LL-Null:                       -5782.2
Covariance Type:            nonrobust   LLR p-value:                     0.000
==================================================================================================
                                    coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
Intercept                        -7.0768      0.286    -24.771      0.000      -7.637      -6.517
Marital_Married[T.True]           0.0730      0.117      0.624      0.533      -0.156       0.302
Marital_Never_Married[T.True]    -0.0391      0.117     -0.335      0.738      -0.268       0.190
Marital_Separated[T.True]         0.1500      0.116      1.297      0.195      -0.077       0.377
Marital_Widowed[T.True]           0.2853      0.115      2.476      0.013       0.059       0.511
Area_Suburban[T.True]            -0.0353      0.091     -0.387      0.699      -0.214       0.144
Area_Urban[T.True]                0.0262      0.091      0.289      0.772      -0.151       0.204
Contract_One_year[T.True]        -3.1580      0.119    -26.499      0.000      -3.392      -2.924
Contract_Two_Year[T.True]        -3.2583      0.117    -27.782      0.000      -3.488      -3.028
Population                    -5.735e-07   2.59e-06     -0.222      0.825   -5.64e-06     4.5e-06
Children                         -0.1133      0.018     -6.208      0.000      -0.149      -0.078
Age                               0.0145      0.002      7.590      0.000       0.011       0.018
Outage_sec_perweek               -0.0006      0.012     -0.048      0.962      -0.025       0.024
Yearly_equip_failure             -0.0187      0.058     -0.321      0.748      -0.133       0.095
Techie                            0.9685      0.098      9.901      0.000       0.777       1.160
Multiple                          0.0914      0.078      1.167      0.243      -0.062       0.245
Tenure                           -0.4527      0.016    -27.840      0.000      -0.485      -0.421
MonthlyCharge                     0.0352      0.001     27.884      0.000       0.033       0.038
Bandwidth_GB_Year                 0.0042      0.000     23.151      0.000       0.004       0.005
Income                         6.655e-07   1.32e-06      0.505      0.613   -1.92e-06    3.25e-06
==================================================================================================
```

```
#Printing AIC, Lower is better!

print(model1.aic)
print(model1.bic)
```

```
4746.6125282146495
4890.819335654173
```

Final Model:

#Model 8, reduce logit model by removing Multiple

#Refit Logit model

model8= logit("Churn ~ Children + Age + Techie + Tenure + MonthlyCharge"

"+ Bandwidth_GB_Year + Contract_One_year + Contract_Two_Year
" ,data=df_model).fit()

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                  Churn   No. Observations:                10000
Model:                          Logit   Df Residuals:                     9991
Method:                           MLE   Df Model:                            8
Date:                Sun, 25 Jan 2026   Pseudo R-squ.:                  0.5920
Time:                        11:43:35   Log-Likelihood:                -2359.3
converged:                       True   LL-Null:                       -5782.2
Covariance Type:            nonrobust   LLR p-value:                     0.000
===============================================================================================
                               coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
Intercept                   -6.9712      0.231    -30.209      0.000      -7.423      -6.519
Contract_One_year[T.True]   -3.1484      0.119    -26.470      0.000      -3.382      -2.915
Contract_Two_Year[T.True]   -3.2403      0.117    -27.753      0.000      -3.469      -3.011
Children                    -0.1137      0.018     -6.246      0.000      -0.149      -0.078
Age                          0.0144      0.002      7.552      0.000       0.011       0.018
Techie                       0.9705      0.097      9.959      0.000       0.780       1.162
Tenure                      -0.4511      0.016    -27.809      0.000      -0.483      -0.419
MonthlyCharge                0.0355      0.001     29.291      0.000       0.033       0.038
Bandwidth_GB_Year            0.0042      0.000     23.114      0.000       0.004       0.005
===============================================================================================
```

```
print(model8.aic)
print(model8.bic)

4736.687050462887
4801.580113810673
```

Conclusion:

The data analysis process began with an initial multiple logistic regression model that included all independent variables identified in question C2. This full model served as the baseline for evaluating the contribution of each predictor in explaining variation in the response variable **Churn**.

A backward stepwise elimination process was applied, justified in question D2 to refine this categorical classification model. This approach iteratively removed predictor variables that were not statistically significant in explaining variation in **Churn**, as indicated by p-values greater than **0.05.** A significance level of 0.05, corresponding to a 95% confidence level, was used as the threshold for variable removal.

The primary model evaluation metrics used to compare iterations of the model were **AIC (Akaike's Information Criteria), BIC (Bayesian Information Criteria), Log-Likelihood, Pseudo R-**

squared, and **LLR p-value. AIC** is prioritized as it evaluates model fit while penalizing unnecessary complexity; lower AIC values indicate a more efficient model.

BIC was used as a secondary confirmation metric, with lower values indicating improved parsimony and goodness of fit.

Log-likelihood values were monitored to assess changes in overall model fit, while Pseudo R-squared was used as a measure of improvement relative to a null(Intercept-Only) model. LLR P-value was used to confirm that each reduced model remained statistically significant compared to the null model.

All five metrics were checked after each removal of a variable for backward stepwise elimination. Across all iterations of variable removal (Outage_sec_perweek, Population, Area, Marital, Yearly_equip_failure, Income, Multiple), **LLR – p-value** stayed the same at **0.00,** indicating an overall highly significant model compared to the null model.

In some iterations of the reduced model, AIC increased slightly but BIC decreased heavily, indicating tradeoff in model5. In model5, where the removal of the one-hot encoded variable "**Marital**" resulted in a decrease Log-likelihood and Pseudo R-squared due to the higher cardinality of the variable and the removal of multiple degrees of freedom. Despite this, the reduction improved overall model parsimony and contributed to a more efficient final model. From the initial model to the reduced model (model8), the AIC and BIC has been transformed from 4746.61, 4890.81 to 4736.68, 4801.58 respectively.

Initial Model confusion Matrix

#Confusion Matrix for model1

conf_matrix_1 = model1.pred_table()

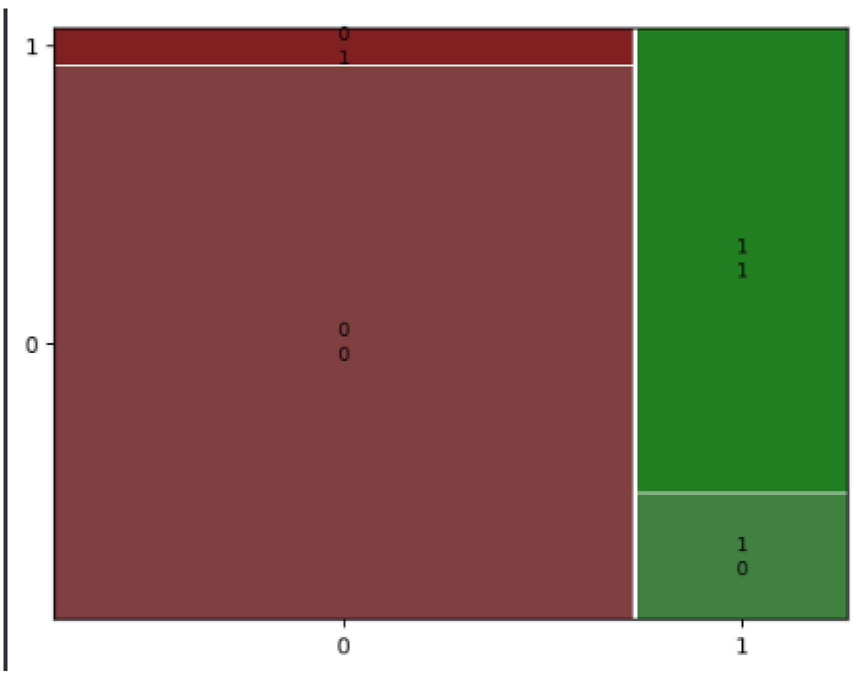
print(conf_matrix_1)

```
#Confusion Matrix for model1

conf_matrix_1 = model1.pred_table()
print(conf_matrix_1)

[[6885.  465.]
 [ 565. 2085.]]
```

mosaic(conf_matrix_1)



#Calculating Accuracy

TN1 = conf_matrix_1[0,0]

TP1 = conf_matrix_1[1,1]

FN1 = conf_matrix_1[1,0]

FP1 = conf_matrix_1[0,1]

#Calculating Accuracy for model1

acc1 = (TN1 + TP1) / (TN1 + TP1 + FN1 + FP1)

print(acc)

```
#Calculating Accuracy for model1

acc1 = (TN1 + TP1) / (TN1 + TP1 + FN1 + FP1)
print(acc)

0.8957
```

Reduced Model Confusion Matrix:

#Creating confusion matrix

conf_matrix = model8.pred_table()

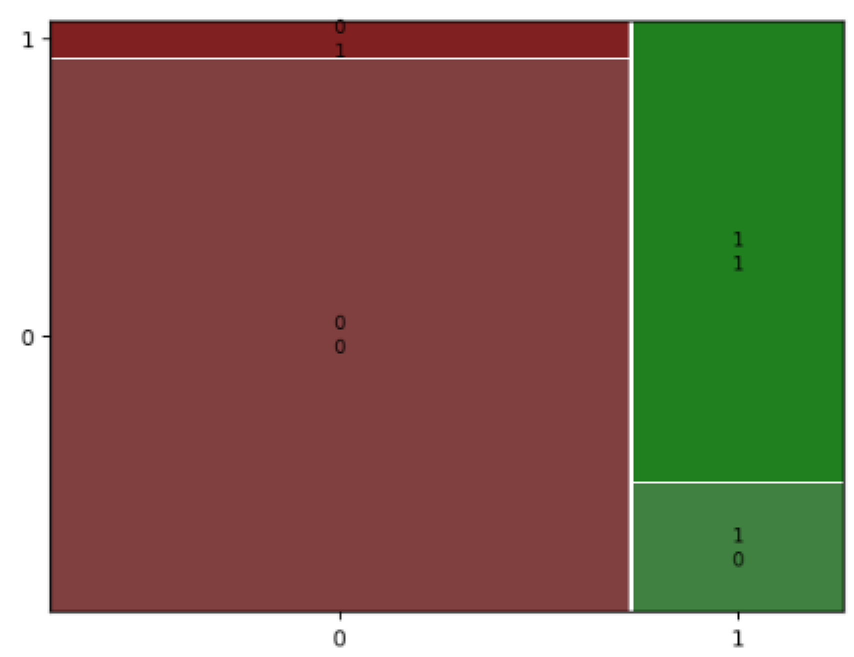
print(conf_matrix)

```
#Creating confusion matrix
conf_matrix = model8.pred_table()

print(conf_matrix)

[[6881.  469.]
 [ 574. 2076.]]
```

mosaic(conf_matrix)



#Calculating Accuracy

TN = conf_matrix[0,0]

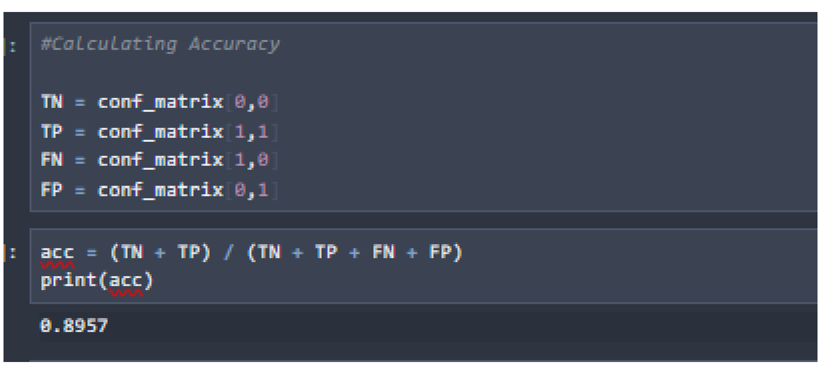
TP = conf_matrix[1,1]

FN = conf_matrix[1,0]

FP = conf_matrix[0,1]

acc = (TN + TP) / (TN + TP + FN + FP)

print(acc)



```
#Calculating Accuracy

TN = conf_matrix[0,0]
TP = conf_matrix[1,1]
FN = conf_matrix[1,0]
FP = conf_matrix[0,1]

acc = (TN + TP) / (TN + TP + FN + FP)
print(acc)

0.8957
```

Interpretation of the coefficients of the reduced model:

Intercept(-6.9712) This represents the estimated baseline Churn for when all predictors are equal to 0 (Contract/Age/Techie etc). While this scenario is unrealistic, the intercept serves as a baseline for other variables to add on to or substract.

Contract_One_year[T.True](-3.1484) Customers on a one year contract are least likely to churn. The negative coefficient indicates an effect suggesting that longer contractual contracts significantly reduce the likelihood of churn.

Contract_Two_Year[T.True](-3.2403) Customers on a two year contract are less likely to churn, even more so than customers with a one year contract. This negative coefficient enforces that longer contracts equates to less likely to churn.

Children(-0.1137) Customers with children and additional children slightly decreases the odds of churn, holding all other variables constant. This suggest customers with more dependents exhibit less churn.

Age(0.0144) As customer ages, chances of churning increases slightly. Indicating that older customers are marginally more likely to churn that younger customers, all else equal.

Techie(0.9705) Customers who consider themselves to be technically inclined in the survey at the time of signup have significantly higher odds of churning than those who are not technically inclined. This suggest that technically inclined customers are more willing to switch providers.

Tenure(-0.4511) Tenure has a strong negative relationship with churn. Implying that long term customers are much more likely to remain with the provider.

MonthlyCharge(0.0355) Higher monthly charges are associated with increased odds of churn. This implies that customers paying more per month are more sensitive to price and are more likely to churn.

Bandwidth_GB_Year(0.0042) Annual bandwidth usage has a positive relationship with churn. Customers with higher data consumption are more likely to churn. Customers with high bandwidth usage possibly demand better performance.

The statistical and practical significance of the reduced model:

The reduced logistic regression model is statistically significant, as indicated by the LLR p-value of 0.00, demonstrating that the retained predictors collectively provide a significantly better fit than the intercept-only (null) model.

Model fit remained strong after backward stepwise elimination. The pseudo R-squared of .5920 indicates substantially explanatory improvement relative to the null model, while reductions in AIC and BIC confirm improved model parsimony without loss of predictive performance. The stability of the LLR p-value across iterations implies that the removed variables did not meaningfully contribute to explaining churn. The model achieved an accuracy of 89.57%, reinforcing that the statistically significant predictors also translate into effective classification performance.

Practically, the reduced model highlights several predictors with strong real-world impact. Contract length showcases the largest effect, with one-year and two-year contracts associated with dramatically lower odds of churn compared to month-to-month contracts. Customer tenure is also negatively associated with churn, suggesting increased retention with long-term customers. Highly monthly charges are associated with higher risk of churn, reflecting price sensitivity. These effects are operationally meaningful and provide insight into actions the company can take to adjust pricing strategy to retain customers.

The limitations of the data analysis:

The logistic regression analysis has several limitations. First, logistic regression models are a classification algorithm for machine learning where it calculates probability of churn rather than the outcome itself, and predicted values are constrained between 0 and 1. Logit also assumes a binomial distribution of the dependent variable. While this is appropriate for binary classification, it limits interpretability compared to linear regression, as coefficients represent changes in log-odds rather than direct changes in the response variable. As a result, coefficients require transformation into odds ratio for practical interpretation, which can be less intuitive for stakeholders.

Second, the analysis assumes a linear relationship between the predictors and the log-odds of churn, which may oversimplify complex customer behavior. Relationships such as pricing sensitivity, tenure effects, or contract decisions may be nonlinear or interact with other variables which are not explicitly modeled in this approach. As logistic regression assumes linearity of independent variables being linearly related to the log odds.

Third, although backward stepwise elimination improved model parsimony, the feature selection process is data-driven and maybe sensitive to the specific sample used. Variables removed due to statistical insignificance may still hold practical relevance.

Additionally the presence of multicollinearity among predictors such as tenure and Bandwidth_GB_Year may amplify coefficients interpretation, even though overall model fit remained strong.

Finally this analysis identifies associations rather than casual relationships. While several predictors show strong statistical and practical significance, results should not be interpreted as casual effects.

Business Recommendation:

Based on the results of the reduced logistic regression model, it is recommended that the company prioritize customer retention strategies focused on incentivizing long term contracts, contract structure, tenure strategies and pricing management, as these variables demonstrated the strongest influence on churn.

Contract length were shown as the most influential predictor of churn. Customers on one-year and two-year contracts were associated with substantially lower odds of churn compared to month-to-month contracts. As a result, the company should incentivize longer-term contracts through targeted promotions, discounts and bundled service offerings to encourage customers who are currently on month-to-month contracts to one/two year contracts.

Customer tenure was also negatively associated with churn, indicating that longer-tenured customers are significantly less likely to leave. This suggest that early stage customers represent the highest risk churners. The company should focus retention efforts on new and short-tenure customers, such as limited time deals, loyalty programs, proactive support reach or early loyalty programs incentives during the first months of service.

MonthlyCharge showed a positive association with churn, implying that customers with higher monthly charges are more likely at risk to churn. This highlights the need for price and value optimization. Targeted discounts, service plan revision, or personalized offers can help mitigate churn risk when it comes to customers with higher than average monthly cost.

Technically inclined customers were positively associated with churn, suggesting that customers who are technically knowledgeable are not satisfied with their service and hold higher expectations that are not being met. It could possibly be performance inconsistencies, service limitations.